

Weather-Aware Load Estimation and Forecasting: Current Practices and Emerging Methods in North America

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Author: David P. Chassin

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Eudoxys Sciences LLC
7443 87th Dr. NE
Marysville, Washington 98270 (USA)

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Executive Summary

Weather is the single most important exogenous driver of electric load in North America, and the growing electrification of space heating and transportation together with the rapid adoption of behind-the-meter (BTM) solar generation are strengthening that dependence. This report reviews weather-aware methods for two closely related problems: *load estimation*, in which load that is unobserved, masked or confounded by weather must be reconstructed or normalized, and *load forecasting*, in which future load is predicted given observed or forecast weather.

The methods presently used by North American utilities, independent system operators (ISOs), regional transmission organizations (RTOs) and planning agencies are reviewed. These include degree-day and temperature-spline regression, statistically adjusted end-use (SAE) models, artificial neural network (ANN) short-term forecasters supplied by multiple weather vendors, weather-scenario simulation for 50/50 and 90/10 peak forecasts, change-point baseline models and weather-sensitive settlement profiles.

This report then examines methods proposed by researchers that appear promising but have not yet been widely adopted, including probabilistic forecasting from numerical weather prediction (NWP) ensembles, gradient-boosted and deep sequence models, hierarchical reconciliation, net-load and BTM photovoltaic (PV) disaggregation, grey-box and physics-informed thermal models, climate-conditioned planning, machine-learning weather models and time-series foundation models. Each is assessed in light of the evidence available from North American data, the constraints of utility operations and the lessons of Winter Storms Uri (2021) and Elliott (2022).

The report concludes with a discussion of the barriers to adoption and a research agenda focused on cold-weather extremes, electrification and calibrated uncertainty.

Contents

1	Introduction	7
2	Scope, Taxonomy, and Terminology	9
2.1	Estimation versus Forecasting	9
2.2	Horizons and Weather Inputs	9
2.3	Spatial Resolution	10
3	Weather Drivers and North American Data Sources	11
3.1	Station Selection and Weighting	11
3.2	Data Products	12
4	Methods in Current Utility and ISO/RTO Practice	13
4.1	Short-Term Operational Forecasting	13
4.2	Medium- and Long-Term Planning	14
4.3	Regulatory Drivers	14
5	Weather-Aware Load Estimation	17
5.1	Weather Normalization of Historical Load	17
5.2	Baselines for Demand Response and M&V	17
5.3	Settlement Load Profiling	17
5.4	Behind-the-Meter PV Estimation and Gross-Load Reconstitution	18
5.5	Distribution Pseudo-Measurements and Load Allocation	18
5.6	Building-Stock and End-Use Load Shape Estimation	18
6	Emerging Research Methods	19
6.1	Probabilistic Forecasting from Weather Ensembles	19
6.2	Additive and Gradient-Boosted Models	19
6.3	Deep Sequence Models	19
6.4	Hierarchical and Spatial Methods	20
6.5	Net-Load, DER-Aware, and Electrification-Aware Models	20
6.6	Physics-Informed and Grey-Box Models	20
6.7	Climate-Conditioned Long-Term Forecasting	20
6.8	Machine-Learning Weather Models	21
6.9	Foundation Models and Transfer Learning	21
7	Evaluation, Benchmarks, and Uncertainty Quantification	23
7.1	Metrics	23
7.2	Public Data Sets and Benchmarks	23
8	Adoption Barriers and Research Agenda	25
8.1	Research Needs	25
9	Conclusions	27

1 Introduction

Load forecasts underpin nearly every decision made in the planning and operation of North American power systems, from day-ahead unit commitment to resource adequacy and transmission plans that extend several decades into the future. Weather accounts for most of the variance in these forecasts. In summer-peaking systems cooling load can exceed 40–50% of peak demand, while in winter-peaking regions such as Québec, Manitoba, the Pacific Northwest and much of the U.S. Southeast the peak is determined largely by resistance and heat-pump heating. The relationship between weather and load has therefore received considerable attention since the classical reviews of load forecasting were written [1–5].

Three developments have made weather awareness more pressing in recent years. The first is the exposure of forecasting and planning shortcomings during extreme weather events. During Winter Storm Uri in February 2021, ERCOT’s planning assumptions understated the cold-weather peak and some 20 GW of firm load was shed at the height of the event [6, 7]. Winter Storm Elliott in December 2022 likewise produced record winter peaks and large forecast errors in PJM, MISO and the Southeast [8]. The second is electrification, which couples new loads to the weather in ways that historical data cannot yet reveal. The output and coefficient of performance of heat pumps decline in cold weather, and the energy required to drive an electric vehicle a given distance increases as temperatures fall [9–11]. The third is the growth of BTM PV, which masks gross load so that the metered quantity becomes a weather-dependent *net* load whose components must be estimated because they cannot be measured directly [12–14]. Taken together these considerations raise questions about whether the methods presently used by utilities and system operators remain adequate, and about which of the many methods proposed in the research literature are ready to replace or supplement them.

The main contributions of this report are (1) a distinction between weather-aware *estimation* and weather-aware *forecasting*, which practitioners often perform with the same models but which must be evaluated differently, (2) a survey of the methods presently used in North American production, drawn from the published methodology manuals of ISOs, RTOs and utilities, (3) a review of research methods that have shown promise on North American data but have not been widely adopted, together with an examination of why this is so, and (4) a discussion of the barriers to adoption and a proposed research agenda. The scope of the review is limited to methods, data and climates relevant to the United States, Canada and the interconnected systems of northern Mexico. Methods developed elsewhere are included only where they have been tested on North American data or can be transferred directly to North American conditions.

The remainder of the report is organized as follows. In Section 2 we present a taxonomy of the problem, and in Section 3 we review the weather variables and data sources available in North America. Section 4 describes the methods in current utility and ISO/RTO practice, and Section 5 addresses weather-aware load estimation. In Section 6 we review emerging research methods, and in Section 7 we discuss evaluation and benchmarking. Finally, in Section 8 we consider the barriers to adoption and propose a research agenda, and Section 9 concludes.

2 Scope, Taxonomy, and Terminology

Weather-aware methods may be organized along three axes, namely the *task* (estimation or forecasting), the *horizon* and the *spatial resolution*. These axes are important because the dominant source of error changes along each of them. At short horizons the error is dominated by the weather forecast itself, whereas at planning horizons it is dominated by the choice of weather scenarios and, increasingly, by the nonstationarity of the climate.

Fig. 1 illustrates how the form of the weather input changes as one moves from estimation through operations to planning, and how the dominant source of error changes with it.

2.1 Estimation versus Forecasting

Weather-aware load estimation infers a load quantity that is unobserved or confounded by weather given weather that has already been observed. It includes the weather normalization of historical load, the construction of counterfactual baselines for demand response (DR) and for the measurement and verification (M&V) of energy efficiency, the reconstitution of gross load masked by BTM PV, the settlement profiling of customers without interval meters, the generation of pseudo-measurements for distribution system state estimation (DSSE) and the calibration of building-stock models. *Weather-aware load forecasting*, by contrast, predicts future load conditioned on forecast weather in the case of operations or on weather scenarios in the case of planning. The same regression model may serve both purposes. However, an estimator is judged on its bias and on the validity of the counterfactual it produces, whereas a forecaster is judged on its out-of-sample accuracy and the calibration of its uncertainty.

2.2 Horizons and Weather Inputs

The conventional forecast horizons, together with their principal users, weather inputs and dominant sources of uncertainty, are summarized in Table 1.

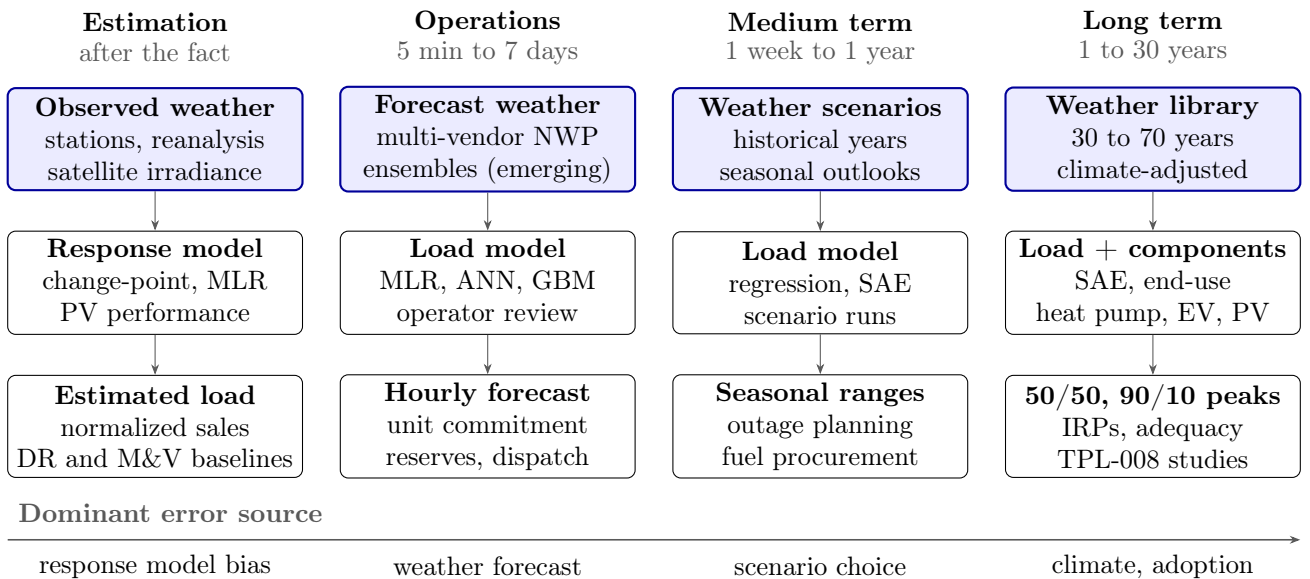


Figure 1: Weather inputs, models and outputs by task and horizon.

Table 1: Forecast Horizons, Users, and Weather Inputs

Horizon	Typical range	Primary users	Weather input	Dominant uncertainty
Very short term (VSTLF)	5 min–2 h	ISO real-time dispatch, AGC	Latest observations, nowcasts, rapid-update NWP (HRRR, RAP)	Model and persistence error
Short term (STLF)	1 h–7 days	Day-ahead markets, unit commitment, utility operations	Deterministic and ensemble NWP from multiple vendors	Weather forecast error, especially fronts and cloud cover
Medium term (MTLF)	1 week–1 year	Outage scheduling, fuel procurement, seasonal assessments	Subseasonal-to-seasonal outlooks, historical weather scenarios	Weather scenario selection
Long term (LTLF)	1–30 years	Resource adequacy, IRPs, transmission planning, rate cases	Normal weather, 30–70 years of historical or climate-adjusted weather	Economic and adoption drivers, climate trend, extreme tails

2.3 Spatial Resolution

Forecasts are produced at the level of the ISO or balancing authority (BA), by weather or load zone (e.g., the eight weather zones used by ERCOT [15]), by substation or feeder for distribution planning and by customer for M&V and settlement. The mapping and weighting of weather stations become more consequential as the resolution increases. In addition, BTM PV and electrification make low-voltage load considerably more sensitive to weather than the system load [16].

Load aggregation is also highly susceptible to the accumulation of errors due to losses in the sub-transmission and distribution systems. Both SCADA and AMI metered load data sources are considered highly accurate. But both technical and non-technical losses can be difficult to account for without additional network topology information, asset condition/status, and accurate records of unaccounted-for energy use and commercial/administrative losses caused by unbilled consumption, electricity theft, faulty meters, or billing system errors.

3 Weather Drivers and North American Data Sources

Temperature remains the dominant driver of load, but its relationship to load is nonlinear, lagged and asymmetric between seasons. Cooling load saturates on very hot days as air-conditioners approach a full duty cycle. Heating load does not saturate in the same way. The FERC–NERC inquiry into Winter Storm Elliott recorded utility testimony that electric heating increases the winter peak nonlinearly as temperatures decrease, and several Southeastern BAs exceeded their 50/50 peak forecasts by 6.8% to 10.3% [8]. As space heating electrifies one should expect this response to become steeper still, since the coefficient of performance (COP) of a heat pump declines with outdoor temperature and backup resistance elements engage at the lowest temperatures [9,10]. The variables most commonly used, the mechanisms by which they affect load and their usual representations in practice are summarized in Table 2.

The selection of weather variables is not universal. Using hourly data from seven U.S. states, Xie and Hong showed that the best weather terms and lag structures differ by region [19,21,24], which suggests that variables should be selected for each zone rather than transferred from one system to another.

3.1 Station Selection and Weighting

Most ISOs and utilities map airport weather stations to zones and form fixed, population-weighted or load-weighted averages. ERCOT’s operational models, for example, use 49 stations [25]. Hong, Wang and White showed that a data-driven selection of the number and combination of stations can reduce forecast error materially when compared to ad hoc selection [26]. Nevertheless, changes in station metadata, instrument outages and urban heat-island effects remain persistent practical problems.

Table 2: Weather Variables, Load Mechanisms, and Their Representation

Variable	Mechanism	Common representation in North American practice	Evidence
Dry-bulb temperature	Conduction and infiltration through building envelopes	Piecewise-linear splines, HDD/CDD with 65°F (U.S.) or 18°C (Canada) base, polynomial terms	[3,17]
Lagged and smoothed temperature	Building thermal mass, occupant behavior, multi-day heat waves and cold snaps	Exponentially weighted temperature, prior-day degree days, “recency” lags	[18]
Humidity and dew point	Latent cooling load, comfort	Temperature–humidity index (THI); PJM uses $THI = DB - 0.55(1 - HUM)(DB - 58)$	[19,20]
Wind speed	Infiltration and convective losses in winter	Wind chill, PJM’s Winter Weather Parameter	[20–22]
Irradiance and cloud cover	Lighting, solar gains, BTM PV output	Cloud-cover indices, GHI from satellite products	[12,23]
Precipitation and snow	Behavior, BTM PV soiling and snow cover	Indicators, mostly in STLF	[3]
Daylight and calendar interaction	Lighting, occupancy	Sunset time, day-of-week $\times$ temperature interactions	[17]

Table 3: Weather Data Products Relevant to North America

Category	Products relevant to North America	Typical use
Station observations	NOAA ISD/ASOS [27]; Environment and Climate Change Canada stations	Model training, settlement, normalization
Reanalysis	ERA5 (1940–present, 0.25°) [28]; MERRA-2 [29]; NARR [30]	Long weather records, gap filling, scenario libraries; ISO-NE uses ERA5 from 1950 [31]
Solar resource	NSRDB [23]; GOES-derived irradiance	BTM PV estimation and net-load models
Typical-year files	TMY3 and successors [32]	Building energy simulation, end-use load shapes
Deterministic NWP	GFS, NAM, RAP [33], HRRR [34]; ECCC GDPS/RDPS/HRDPS; ECMWF IFS	STLF and VSTLF inputs
Ensemble NWP and post-processed blends	GEFS [35], ECMWF ENS, ECCC GEPS; NOAA National Blend of Models	Probabilistic STLF, scenario generation
Machine-learning weather models	GraphCast [36], GenCast [37], Pangu-Weather [38], ECMWF AIFS [39]	Emerging; see Section 6
Downscaled climate projections	CMIP6 downscaled with LOCA2 [40]; climate-adjusted ERA5	Climate-conditioned LTLF; the California Energy Commission and ISO-NE use these

3.2 Data Products

The principal sources of weather data used for load estimation and forecasting in North America are listed in Table 3.

Operational forecasters rarely rely on a single weather source. ERCOT, for instance, runs its internal models separately on each of four global models (GFS, GEFS, ECMWF and NAM) and three vendor forecasts, and has planned to expand to 14 weather inputs [25].

4 Methods in Current Utility and ISO/RTO Practice

Load forecasting practice in North America is conservative, transparent and centered on regression. Neural networks have been used in production for short-term forecasting since the 1990s, but planning forecasts that are filed in regulatory proceedings continue to rely on models that the commissions can audit. The principal change over the past five years has not been in the core models but in the treatment of weather. Utilities and system operators now use more weather sources at short horizons, longer and sometimes climate-adjusted weather libraries at long horizons, and explicit weather-sensitive components for PV, heat pumps and electric vehicles.

4.1 Short-Term Operational Forecasting

1. **Multiple linear regression (MLR) with weather interactions.** Hourly load is regressed on polynomials or splines of temperature interacted with the hour of day, the day of week and the month, together with lagged temperature and holiday terms [17, 41]. The “vanilla” model of Hong *et al.* [17] served as the benchmark for the Global Energy Forecasting Competitions [42, 43] and is close to many production specifications.
2. **Artificial neural networks.** EPRI’s ANNSTLF, which is built on separate base-load and load-change networks with weather inputs, was widely deployed by North American utilities [44]. ERCOT’s internal mid-term forecaster combines neural network and linear regression models and runs a separate instance for each weather source [25]. Many utilities use commercial platforms that combine regression and ANN components with a weather-feed manager.
3. **Similar-day and pattern-matching methods.** Operators identify historical days with similar weather and calendar attributes and adjust them for load growth. These methods remain in use as operator cross-checks, particularly for holidays and unusual events.
4. **Multi-source weather blending and forecaster judgment.** ISOs ingest several NWP and vendor feeds and combine the resulting load forecasts. MISO receives hourly vendor forecasts of temperature, dew point, cloud cover and precipitation and refreshes its 168-hour medium-term forecast every 15 minutes, and its meteorologists override inputs they judge to be unreliable [45]. Manual adjustment during frontal passages, convective events and cold snaps remains common, although it is rarely documented quantitatively.
5. **Very short-term extrapolation.** ERCOT’s short-term forecast produces 5-minute values for the next two hours [25]. Models at this horizon are typically autoregressive or persistence-plus-ramp models in which weather enters mainly through recent observations.
6. **Gross-load reconstitution for BTM PV.** In systems with significant distributed PV, ISOs estimate BTM PV output from installed capacity and irradiance, add it back to metered load, forecast the resulting gross load and then subtract the forecast PV output [12, 15, 31]. This separates the sensitivity of PV output to cloud cover from the sensitivity of demand to temperature.

4.2 Medium- and Long-Term Planning

1. **Weather-normalized econometric models.** Annual or monthly energy is regressed on economic drivers and on heating and cooling degree days, and the result is evaluated at “normal” weather, which is usually a 30-year average such as the NOAA 1991–2020 normals. Peak models use the weather on the peak day (e.g., THI or a weighted multi-day temperature) and are evaluated at a design condition. NYISO’s transmission owners, for example, use design criteria with an annual probability of exceedance no greater than 0.5 [46].
2. **Statistically adjusted end-use (SAE) models.** The SAE models developed and popularized by Itron construct heating, cooling and base-use variables as products of end-use saturation, efficiency and building-stock indices with degree days, and then estimate their coefficients by regression [47]. They thereby connect equipment efficiency standards and appliance saturation to the weather response, and they are widely used in utility integrated resource plans (IRPs) and by ISOs. ISO-NE has published an introduction to SAE modeling for its stakeholders [47].
3. **End-use simulation models.** Ontario’s IESO uses an End-Use Forecaster across ten zones with modules for equipment stock, equipment choice and customer growth, and converts annual sectoral energy to hourly load using end-use load shapes [48]. The demand analysis of the California Energy Commission follows a similar end-use tradition.
4. **Weather-scenario simulation.** Rather than a single normal year, the fitted model is driven by many historical weather years and the 50/50 and 90/10 forecasts are defined by percentiles of the simulated seasonal peaks. Practice varies widely (Table 4). PJM rotates historical weather through its zonal regression models [20]. ERCOT applies 17 weather years from 2008 to 2024 [15]. IESO runs 31 years (1993–2023) shifted by ± 7 days to create 465 hourly simulations [48]. ISO-NE now simulates all hours for all load components across 70 years of ERA5 weather, adjusted using ten scenarios from five climate models under two emissions pathways [31]. The statistical basis for this approach is the density-forecasting framework of Hyndman and Fan [49] and the hourly long-term probabilistic method of Hong, Wilson and Xie [50].
5. **Weather-sensitive component models.** ISO-NE forecasts heat-pump load from simulated building heating requirements and COP curves, with heating triggered below 62°F, and forecasts EV load with temperature-sensitive efficiency [31]. BTM PV adoption is modeled with NREL’s dGen model and converted to hourly output using historical capacity-factor profiles [31]. These components are then added to the base load in a fixed order of attribution.

4.3 Regulatory Drivers

Two regulatory developments are encouraging better treatment of weather. Following Uri and Elliott, FERC and NERC recommended that BAs improve their short-term forecasts for periods of extreme cold and share effective practices with one another [6, 8]. In addition, NERC’s extreme-temperature transmission planning standard (TPL-008) requires planners to study benchmark extreme heat and cold events, which in turn requires load forecasts conditioned on those events. Each benchmark must be drawn from at least 40 years of temperature data and rank among the

Table 4: Weather treatment in selected North American planning forecasts (as published 2025–2026)

Organization	Core load model	Weather library	50/50 and 90/10 construction	Electrification and PV treatment	Source
ISO-NE	Daily linear regression feeding 24 hourly neural networks; ~25 weather and 40+ calendar features	70 years of ERA5, climate-adjusted with 10 scenarios	Percentiles of 70 simulated seasonal peaks	Heat pumps via COP curves; EVs with temperature-sensitive efficiency; BTM PV via dGen	[31]
PJM	Zonal multiple regression with THI and Winter Weather Parameter	Historical weather rotation	Median and 10th/90th percentiles of simulated peaks	Distributed solar and storage adjustments; large-load adjustment process; 20-year horizon	[20]
ERCOT	Linear regression on weather, premise counts, calendar; 8 weather zones	17 historical years (2008–2024)	Mean of scenarios (P50); P90 of scenarios	BTM PV reconstitution; EV registrations; large-load realization factors	[15]
IESO (Ontario)	End-Use Forecaster plus demand simulation weather model	31 years (1993–2023), ± 7 -day shifts, 465 simulations	Month-by-month median selection for normal year	Separate stock-and-flow EV model	[48]
NYISO	Transmission-owner regression forecasts reviewed by NYISO	Owner-specific	Design criteria ≤ 0.5 annual exceedance probability	Owner-specific	[46]

20 most extreme three-day rolling-average events [51]. NERC’s most recent long-term reliability assessment also emphasizes winter risk and load growth driven by electrification across most assessment areas [52].

5 Weather-Aware Load Estimation

The estimation problems discussed in this section share a common structure. A weather-response model is fitted where load is observed and then evaluated where it is not, whether at counterfactual weather, in a counterfactual period or for an unobserved component. The quality of the estimate therefore depends on whether the model is causally adequate and not merely on whether it is predictive. A baseline that forecasts well but absorbs part of the effect of a DR event into a temperature term is, for example, a poor estimator of that effect.

5.1 Weather Normalization of Historical Load

Utilities normalize sales for rate cases, revenue reporting and model calibration by fitting load to heating and cooling degree days and then re-evaluating the fit at normal weather. The PRISM method, which fits a heating (or cooling) change-point at a household-specific balance temperature, remains the conceptual basis for most normalization of billing data [53]. The normalization of peaks is more difficult because the peak-day weather of any single year is a noisy draw, and simulation over long weather histories (Section 7) is increasingly used for both energy and peak.

5.2 Baselines for Demand Response and M&V

1. **Averaging baselines with weather adjustment.** ISO and RTO DR programs commonly use “X-of-Y” day averages with a same-day additive or multiplicative adjustment, and sometimes offer a weather-sensitive regression baseline as an alternative. Coughlin *et al.* found that a same-day (morning) adjustment improved the accuracy of most baseline models for non-residential buildings [54].
2. **Regression baselines.** Mathieu *et al.* proposed a regression on time of week and outdoor temperature with piecewise-linear temperature segments, which has since been widely adopted for commercial buildings [55].
3. **Change-point models.** ASHRAE Guideline 14 specifies two- to five-parameter change-point models together with the uncertainty statistics CV(RMSE) and NMBE used to accept them for M&V [56], and these models are implemented in the Inverse Modeling Toolkit [57]. The CalTRACK methods used for pay-for-performance programs in California standardize daily and hourly degree-day change-point models on AMI data [58].

5.3 Settlement Load Profiling

Where interval data are unavailable, retail settlement relies on assigned load profiles. ERCOT classifies each service point by profile type, weather zone, meter data type, weather sensitivity and time-of-use schedule, and backcasts its weather-sensitive profiles using NOAA station data [59]. Separate profile segments exist for PV and other distributed generation. The deployment of AMI has reduced the role of profiling, but it remains important in regions and customer classes that still have legacy meters.

5.4 Behind-the-Meter PV Estimation and Gross-Load Reconstitution

BTM PV is rarely metered separately, so its output must be inferred. Three approaches are in use or close to it.

1. **Model-based upscaling.** Installed capacity from interconnection records is combined with irradiance-driven performance models using satellite irradiance data such as the NSRDB [23]. This is the standard ISO approach [15, 31].
2. **Representative-site upscaling.** Shaker, Zareipour and Wood estimated the output of “invisible” sites from a small number of metered sites and publicly available weather data [13].
3. **Source separation.** Contextually supervised source separation decomposes net load into weather-driven solar and consumption components using irradiance and temperature as context [60]. Wang *et al.* showed that probabilistic net-load forecasts improve when BTM PV is modeled explicitly [12].

5.5 Distribution Pseudo-Measurements and Load Allocation

Distribution planners allocate measured feeder-head load to transformers in proportion to billed energy or transformer kVA, and DSSE requires pseudo-measurements at nodes that are not monitored. Weather-driven regression and ANN models for pseudo-measurements reduce DSSE error when compared to static load shapes [61]. AMI data now permit thermal response to be estimated at the customer level. Albert and Rajagopal [62] and Kwac, Flora and Rajagopal [63] used smart-meter data from Pacific Gas and Electric to estimate household temperature sensitivity and to segment customers by their load-shape response, and a broader review of such methods is given in [64]. Nonintrusive load monitoring [65] can separate HVAC from other end uses, but at AMI resolution a weather-regression decomposition is usually more robust.

5.6 Building-Stock and End-Use Load Shape Estimation

NREL’s End-Use Load Profiles project calibrated the ResStock and ComStock building-stock models against utility AMI and end-use metering data across the United States, and publishes end-use load shapes for both typical and actual meteorological years [66]. These models are built on EnergyPlus [67]. Agent-based distribution simulators such as GridLAB-D attach physics-based thermostatic load models to weather files, which produces weather-consistent estimates of feeder load and of DR response [68, 69]. These tools have become the standard source of weather-sensitive end-use load shapes for electrification studies, although utilities generally recalibrate them to local conditions.

6 Emerging Research Methods

This section reviews methods for which there is credible evidence on North American data but which are not yet widely used in production. Their maturity and the principal obstacles to their adoption are summarized in Table 5.

6.1 Probabilistic Forecasting from Weather Ensembles

Production STLF continues to treat weather as a deterministic input, even when several weather feeds are blended. Yet Taylor and Buizza showed two decades ago that propagating the 51 members of an NWP ensemble through a load model yields better-calibrated load densities, and more accurate point forecasts, than a single weather forecast at lead times beyond about one day [70,71]. The meteorological community has since developed post-processing methods that correct the bias and dispersion of ensembles. These include ensemble model output statistics (EMOS) [72], Bayesian model averaging [73], ensemble copula coupling to restore spatial and temporal dependence [74] and neural-network post-processing [75]. For medium and long horizons, Xie and Hong generated temperature scenarios by shifting historical weather by days and years, and showed how the number of scenarios affects the accuracy of the resulting quantiles [76]. The necessary building blocks therefore exist. What is missing in practice is an end-to-end calibration from post-processed ensemble weather through to load quantiles that operators would trust for the procurement of reserves.

Quantile regression averaging over “sister” forecasts (i.e., point models that differ only in their selection of variables) offers a simple path to probabilistic output from existing MLR models [77]. The GEFCom2014 and GEFCom2017 competitions, the latter of which used ISO-NE zonal data, showed that weather scenario generation combined with quantile regression or GAMs is consistently competitive [43,78].

6.2 Additive and Gradient-Boosted Models

Generalized additive models (GAMs) represent temperature effects as smooth functions and interactions as tensor-product smooths, and they fit well with the regression culture of planning organizations [79,80]. Gradient-boosted trees captured the nonlinear interactions of temperature, lags and calendar variables in GEFCom2012 using U.S. utility data [81,82], and implementations such as XGBoost [83] are now common in research and in some vendor products. Both families of models can be made interpretable enough for regulatory use with partial-dependence plots and SHAP values. Nevertheless, planners have been slow to replace SAE models with either.

6.3 Deep Sequence Models

Recurrent networks such as LSTMs have improved residential load forecasts relative to feed-forward networks [84]. Global sequence models trained across many series perform well on benchmark data sets, including DeepAR [85], the Temporal Fusion Transformer (TFT) with explicit known-future weather covariates [86] and N-BEATS adapted for mid-term load [87]. Bayesian deep learning provides estimates of net-load uncertainty at the customer level [14]. The evidence for these models is strongest at the customer and feeder levels, where load is noisy and many series are available. At the system level the gains over well-specified MLR and GBM models are typically modest. Moreover, their performance degrades in rare weather conditions that are

under-represented in the training data, which are precisely the conditions in which operators most need accurate forecasts.

6.4 Hierarchical and Spatial Methods

Forecasts at the customer, feeder, zonal and system levels rarely add up. Optimal reconciliation methods [88,89] and their probabilistic extension to smart-meter hierarchies [90] produce coherent forecasts and often improve accuracy at every level. GEFCom2017 framed hierarchical probabilistic load forecasting using ISO-NE data [78]. Spatial approaches that learn the weighting of stations into zones, including graph neural networks over station networks, generalize the station-selection problem of [26]. Few North American utilities reconcile forecasts across hierarchies today, in part because distribution and transmission forecasts are the responsibility of different groups.

6.5 Net-Load, DER-Aware, and Electrification-Aware Models

As PV, storage, heat pumps and electric vehicles grow, the weather response of load changes faster than historical data can reveal. Research approaches to this problem include joint probabilistic models of consumption and BTM PV [12,14], component models that drive heat-pump load with COP curves and temperature [9,10] and EV charging models with temperature-dependent energy use [11,91]. Waite and Modi estimated that full electrification of space heating in New York would shift the state to a winter peak substantially higher than its present summer peak [9], and White *et al.* found a similar risk of large winter peaks in ERCOT under residential heating electrification [10]. ISO-NE has adopted component models of this kind in its planning [31]. However, most utilities still add electrification as an annual energy adjustment with a fixed load shape, which fails to capture the steepening of cold-weather peaks.

6.6 Physics-Informed and Grey-Box Models

Grey-box methods embed thermal physics in statistical models. Aggregate thermostatic load models represent populations of heating and cooling devices as coupled thermal states [69,92], and RC-network building models link outdoor temperature, solar gains and thermal mass explicitly. Physics-informed neural networks add physical residuals to the training loss [93]. These approaches promise better extrapolation to unprecedented cold and heat, DR baselines that are physically consistent and a natural means of incorporating building-stock data such as [66]. But the evidence at system scale remains limited, and the identification of model parameters from aggregate load is a poorly conditioned problem.

6.7 Climate-Conditioned Long-Term Forecasting

Using past weather as a sample of future weather understates the effect of warming and may misstate the tails of the distribution. Auffhammer, Baylis and Hausman estimated that climate change will increase the frequency and intensity of peak demand across U.S. balancing authorities [94], and Burillo *et al.* projected increases in peak demand for Los Angeles under higher temperatures [95]. Downscaled CMIP6 products such as LOCA2 [40] supply station-level temperature trajectories. The California Energy Commission uses hourly station projections derived from LOCA2 [96], and ISO-NE climate-adjusts 70 years of ERA5 [31]. Several questions remain

Table 5: Maturity of emerging methods for North American weather-aware load forecasting

Method family	Evidence on North American data	Adoption status	Main barrier
Ensemble-driven probabilistic STLF	Moderate; foundational studies used European data [70, 71]	Pilots; multi-feed deterministic blending is common	Calibration, data volume, operator trust
Quantile regression and GAMs	Strong (GEFCom)	Limited in planning, some vendors	Retraining of staff and regulators
Gradient-boosted trees	Strong	Growing in vendor products	Extrapolation to unseen weather
Deep sequence models	Strong at customer level, mixed at system level	Research and some vendors	Data needs, interpretability, tail performance
Hierarchical reconciliation	Moderate	Rare	Organizational silos
Electrification component models	Moderate	Adopted by ISO-NE; rare elsewhere	Lack of metered heat-pump and EV data
Grey-box and physics-informed	Emerging	Research	Identifiability, scale
Climate-conditioned planning	Moderate	ISO-NE, CEC	Downscaling choices, regulatory acceptance
ML weather models as inputs	Very limited	None documented	Validation for extremes and surface variables
Foundation models	Emerging (BuildingsBench)	Research	Weak use of weather covariates

open, including how historical weather can be adjusted without destroying its spatial and temporal structure, how changes in cold extremes (which are not simply warming shifts) should be represented and how climate scenarios can be combined with the TPL-008 benchmark events [51].

6.8 Machine-Learning Weather Models

Data-driven weather models trained on ERA5 now match or exceed physics-based NWP on many medium-range metrics. GraphCast [36] and Pangu-Weather [38] are deterministic, GenCast [37] produces probabilistic ensembles and ECMWF’s AIFS [39] has entered operational use. The low inference cost of these models makes large ensembles affordable, which could make the ensemble-to-load pipeline described in Section 6 practical. The risks for load forecasting include the smoothing of extremes, limited skill near the surface and in the boundary layer, and a resolution that is coarse relative to station-level load models. We found no published evaluation of these models as inputs to operational North American load forecasts, which is a clear gap in the literature.

6.9 Foundation Models and Transfer Learning

Pre-trained time-series foundation models, including Chronos [97] and TimesFM [98], forecast zero-shot from history alone. BuildingsBench, which is built on simulated U.S. building-stock load, showed that pre-training on synthetic buildings transfers to real commercial and residential buildings [99]. For weather-aware forecasting the principal limitation is that many foundation models ignore or make weak use of exogenous covariates, whereas weather is the main driver of

load. Transfer learning appears most promising for new service areas, newly electrified loads and data-poor co-operatives and municipal utilities.

7 Evaluation, Benchmarks, and Uncertainty Quantification

A weather-aware forecast has two sources of error, namely the load model and the weather forecast or scenario that drives it. Evaluations should therefore report both the *ex post* error, computed with observed weather, and the *ex ante* error, computed with the weather forecasts available at the time of issue [3]. Without this distinction a better weather vendor can be mistaken for a better load model, and vice versa.

7.1 Metrics

MAPE remains the metric most often reported by industry, but it obscures performance at the peak and is unsuitable for net load near zero. Probabilistic forecasts should be evaluated with proper scoring rules such as the pinball (quantile) loss and the continuous ranked probability score (CRPS), together with calibration diagnostics such as probability integral transform histograms and quantile coverage [100]. Because the most costly errors occur in extreme weather, we recommend that studies also report error conditioned on the weather regime (e.g., the coldest and hottest 1% of hours) and threshold-weighted CRPS for peak conditions. Conformal prediction offers distribution-free coverage guarantees, and its adaptive variants address the distribution shift caused by electrification and climate change [101].

7.2 Public Data Sets and Benchmarks

The public data sets most relevant to North American studies are listed in Table 6.

Table 6: Public Data Sets and Benchmarks

Data set	Content	North American relevance
GEFCom2012 [42]	Hourly load for 20 zones of a U.S. utility with 11 weather stations	Station-selection and hierarchical benchmarks
GEFCom2014 [43]	Hourly load and 25 stations, probabilistic track	Probabilistic benchmark
GEFCom2017 [78]	ISO-NE zonal load and weather	Hierarchical probabilistic benchmark
EIA Hourly Electric Grid Monitor (Form EIA-930) [102]	Hourly demand and day-ahead demand forecasts for U.S. BAs	Enables cross-BA comparison against operational forecasts
NREL End-Use Load Profiles [66]	Simulated, calibrated end-use load shapes for U.S. building stock	Electrification and estimation studies
BuildingsBench [99]	900K simulated buildings plus real building data	Pre-training and transfer learning

The EIA-930 data are particularly under-used. Because they include each BA’s own day-ahead forecast, they allow new methods to be evaluated directly against North American production practice, including during Uri, Elliott and subsequent cold events.

8 Adoption Barriers and Research Agenda

Promising methods are not always adopted by utilities and regulator for the following reasons.

1. **Regulatory auditability.** Planning forecasts are litigated in rate cases and IRP proceedings. Commissions and intervenors can audit an SAE regression, but they cannot easily audit a deep neural network.
2. **Rare-event validation.** The events that matter most, such as Uri and Elliott, represent only a handful of days in a training record. A model that performs best on average error may perform poorly on those days, and current studies rarely test for this.
3. **Data access.** AMI data are held by distribution utilities under privacy constraints, whereas bulk system forecasts are made by ISOs. In addition, heat-pump, EV and BTM PV output are seldom metered separately.
4. **Organizational separation.** Operations, planning, distribution and DR groups often maintain separate models, weather feeds and station mappings, which prevents consistency across hierarchies and horizons.
5. **Vendor dependence.** Many utilities rely on commercial forecasting platforms, so that adoption depends on the product roadmaps of vendors.
6. **Nonstationarity.** Electrification, PV growth, data centers and climate change all make historical weather-response relationships less reliable, which undermines the validation logic on which every data-driven method depends.

8.1 Research Needs

As a result of these concerns the following research needs have been identified.

1. **A cold-weather extreme benchmark.** BA-level load, operational forecasts, NWP ensembles and outage data for Uri, Elliott and subsequent cold events should be assembled so that methods can be compared on the tail of the distribution and not only on the mean.
2. **Operational trials of ensemble-to-load pipelines.** Post-processed physics-based and ML weather ensembles [37, 39] should be tested as inputs to probabilistic STLF, and the resulting quantiles connected to reserve and commitment decisions.
3. **Electrification-aware gross-load reconstitution.** BTM PV reconstitution should be extended to heat pumps and EVs using utility pilots, connected-thermostat data and EULP-style simulations [66].
4. **Standards for climate-adjusted weather libraries.** Shared and documented methods are needed for adjusting historical weather to current and future climates, including cold extremes, consistent with the TPL-008 benchmark events [51].
5. **Hybrid physics–ML models for extrapolation.** Grey-box thermal structure should be combined with flexible learned residuals and the resulting models tested specifically on out-of-sample extreme temperatures.

6. **Coherent transmission–distribution forecasts.** Customer, feeder, zonal and system forecasts should be reconciled probabilistically [90] using common weather inputs.
7. **Covariate-aware foundation models.** Pre-training on large building-stock simulations should treat weather as a first-class input, and the resulting models should be evaluated on EIA-930 data against production forecasts.
8. **Interpretable reporting for regulators.** Disclosure templates (e.g., weather-sensitivity curves, scenario libraries and tail metrics) should be developed so that commissions can review machine-learning forecasts in the same way they review SAE models today.

9 Conclusions

North American load forecasting practice performs well near the center of the distribution and poorly in its tails. Regression, SAE and ANN models with multi-source weather inputs achieve low average error, and the leading planning organizations have moved from a single normal year to large weather libraries, with ISO-NE now climate-adjusting 70 years of reanalysis data. However, the failures during Uri and Elliott occurred in cold extremes where heating load does not saturate and where electrification will steepen the response further.

The research methods most likely to close this gap are not necessarily the most complex. Ensemble-driven probabilistic forecasting, electrification-aware component models, climate-adjusted weather libraries and grey-box hybrids each address a specific failure mode and can each be audited. Deep learning, machine-learning weather models and foundation models are promising, but they require validation on extreme North American events and reporting conventions that regulators can review. It is our view that a shared cold-weather benchmark built on EIA-930 and post-event data would be the most useful single step toward their adoption.

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Appendix A - Acronyms

AGC	Automatic generation control
AIFS	Artificial Intelligence Forecasting System (ECMWF)
AMI	Advanced metering infrastructure
ANN	Artificial neural network
ANNSTLF	Artificial Neural Network Short-Term Load Forecaster (EPRI)
ASHRAE	American Society of Heating, Refrigerating and Air-Conditioning Engineers
ASOS	Automated Surface Observing System
BA	Balancing authority
BTM	Behind-the-meter
CDD	Cooling degree days
CEC	California Energy Commission
CELT	Capacity, Energy, Loads, and Transmission (ISO-NE annual report)
CMIP6	Coupled Model Intercomparison Project, Phase 6
COP	Coefficient of performance
CRPS	Continuous ranked probability score
CV(RMSE)	Coefficient of variation of the root-mean-square error
DB	Dry-bulb temperature (in the PJM THI formula)
DER	Distributed energy resources
DR	Demand response
DSSE	Distribution system state estimation
ECCC	Environment and Climate Change Canada
ECMWF	European Centre for Medium-Range Weather Forecasts
EIA	U.S. Energy Information Administration
EMOS	Ensemble model output statistics
ENS	ECMWF ensemble forecast
EPRI	Electric Power Research Institute
ERA5	Fifth-generation ECMWF atmospheric reanalysis
ERCOT	Electric Reliability Council of Texas
EULP	End-Use Load Profiles (NREL project)
EV	Electric vehicle
FERC	Federal Energy Regulatory Commission
GAM	Generalized additive model
GBM	Gradient-boosting machine
GDPS	Global Deterministic Prediction System (ECCC)
GEFCom	Global Energy Forecasting Competition
GEFS	Global Ensemble Forecast System (NOAA)
GEPS	Global Ensemble Prediction System (ECCC)

GFS Global Forecast System (NOAA)

GHI Global horizontal irradiance

GOES Geostationary Operational Environmental Satellite

HDD Heating degree days

HRDPS High Resolution Deterministic Prediction System (ECCC)

HRRR High-Resolution Rapid Refresh

HUM Relative humidity (in the PJM THI formula)

HVAC Heating, ventilation, and air conditioning

IESO Independent Electricity System Operator (Ontario)

IFS Integrated Forecasting System (ECMWF)

IRP Integrated resource plan

ISD Integrated Surface Database (NOAA)

ISO Independent system operator

ISO-NE ISO New England

LOCA2 Localized Constructed Analogs, version 2

LSTM Long short-term memory

LTLF Long-term load forecasting

M&V Measurement and verification

MAPE Mean absolute percentage error

MERRA-2 Modern-Era Retrospective Analysis for Research and Applications, Version 2

MISO Midcontinent Independent System Operator

ML Machine learning

MLR Multiple linear regression

MTLF Medium-term load forecasting (also ERCOT’s Mid-Term Load Forecast)

N-BEATS Neural basis expansion analysis for time series

NAM North American Mesoscale Forecast System

NARR North American Regional Reanalysis

NERC North American Electric Reliability Corporation

NMBE Normalized mean bias error

NOAA National Oceanic and Atmospheric Administration

NREL National Renewable Energy Laboratory

NSRDB National Solar Radiation Database

NWP Numerical weather prediction

NYISO New York Independent System Operator

PJM PJM Interconnection

PRISM Princeton Scorekeeping Method

PV Photovoltaic

RAP Rapid Refresh

RC Resistance–capacitance (thermal network model)
RDPS Regional Deterministic Prediction System (ECCC)
RMSE Root-mean-square error
RTO Regional transmission organization
SAE Statistically adjusted end-use
SCADA Supervisory control and data acquisition
SHAP SHapley Additive exPlanations
STLF Short-term load forecasting
TFT Temporal Fusion Transformer
THI Temperature–humidity index
TMY3 Typical Meteorological Year, version 3
TPL-008 NERC reliability standard for transmission planning in extreme temperature events
VSTLF Very short-term load forecasting

Appendix B - Open-Source Code Repositories

The repositories listed below implement the methods discussed in this report or provide access to the data on which those methods depend. The existence and active status of each repository were confirmed on October 1, 2026. Licenses are given as declared on GitHub, and “custom” denotes non-standard terms that should be reviewed before reuse. A number of former NREL repositories now redirect to the NatLabRockies organization on GitHub, and the current paths are shown.

Note that no public code exists for several of the production methods described in this report, including Itron’s SAE models and MetrixND, EPRI’s ANNSTLF, the operational forecasters of the ISOs and RTOs, ISO-NE’s CELT models and IESO’s End-Use Forecaster.

NWP data access

- `blaylockbk/Herbie` (Python). Downloads HRRR, RAP, GFS, IFS, and other NWP output from NOAA, cloud archives, and ECMWF open data; the practical route to the inputs in [33–35].

Reanalysis access

- `ecmwf/cdsapi` (Python). Client for the Copernicus Climate Data Store, the source of ERA5 [28].

Solar resource access

- `NatLabRockies/rex` (Python). Resource extraction tool for NSRDB and other NREL resource data sets [23].

ISO/RTO data access

- `gridstatus/gridstatus` (Python). Retrieves data from North American ISOs and other grid sources; useful for benchmarking against operational forecasts.

Regression and hierarchical forecasting

- `tidyverts/fable` (R). Tidy forecasting framework with regression, ARIMA, and ETS models; the `fabletools` companion supports reconciliation.

Hierarchical forecasting

- `earowang/hts` (R). Implements the optimal combination method of [88]; superseded by `fable` for new work.

Probabilistic reconciliation

- `Nixtla/hierarchicalforecast` (Python, Apache-2.0). MinT [89] and probabilistic reconciliation methods.

GAMs

- `dswah/pyGAM` (Python, Apache-2.0). Generalized additive models in Python; the R package `mgcv`, used in [79, 80], is distributed through CRAN.

Gradient boosting

- `dmlc/xgboost` (C++/Python/R, Apache-2.0). Reference implementation of [83].
- `lightgbm-org/LightGBM` (C++/Python/R, MIT). Fast histogram-based gradient boosting; formerly hosted under Microsoft.

Model interpretation

- `shap/shap` (Python). SHAP values for explaining tree and neural models to regulators.

Deep probabilistic models

- `awslabs/gluonts` (Python, Apache-2.0). Includes the DeepAR implementation of [85].

Deep sequence models

- `sktime/pytorch-forecasting` (Python, MIT). Temporal Fusion Transformer [86] with known-future co-variates such as weather forecasts.

- `Nixtla/neuralforecast` (Python, Apache-2.0). N-BEATS, TFT, LSTM, and other models with exogenous inputs.
- `unit8co/darts` (Python, Apache-2.0). Unified interface to statistical, boosted, and deep models with past and future covariates.

N-BEATS

- `ServiceNow/N-BEATS` (Python, custom). Original N-BEATS code; not updated since 2022.

Ensemble post-processing

- `slerch/ppnn` (Python/R). Code for the neural-network post-processing experiments of [75].

Proper scoring rules

- `frazane/scoringrules` (Python, Apache-2.0). CRPS and other proper scores [100] for probabilistic load forecasts.

Conformal prediction

- `scikit-learn-contrib/MAPIE` (Python, BSD-3-Clause). Conformal prediction intervals, including methods for time series [101].

ML weather models

- `google-deepmind/weathernext` (Python, Apache-2.0). GraphCast [36] and GenCast [37]; renamed from graphcast. Model weights carry separate terms.
- `198808xc/Pangu-Weather` (Python, none declared). Official Pangu-Weather inference code [38].
- `ecmwf/anemoui-core` (Python, Apache-2.0). Core packages of ECMWF’s Anemoui framework, used to build AIFS [39].

Foundation models

- `amazon-science/chronos-forecasting` (Python, Apache-2.0). Chronos pretrained models [97].
- `google-research/timesfm` (Python, Apache-2.0). TimesFM pretrained model [98].

Foundation models

- `NatLabRockies/BuildingsBench` (Python, BSD-3-Clause). Pre-training and benchmarking code for [99].

M&V baselines

- `opensm/opensm` (Python, Apache-2.0). Successor to OpenEEmeter; implements CalTRACK daily and hourly baselines [58].
- `opensm/caltrack` (CC0-1.0). CalTRACK methods documentation and testing [58].
- `LBNL-ETA/RMV2.0` (R). LBNL tool for advanced M&V using interval meter data, in the tradition of [55].
- `kW-Labs/nmecr` (R). Site-specific M&V algorithms, including change-point models of the type in [56, 57].

Load disaggregation

- `nilmtk/nilmtk` (Python). Non-intrusive load monitoring toolkit [65].

BTM PV estimation

- `pvlib/pvlib-python` (Python). PV performance modeling from irradiance and temperature, for model-based upscaling.

BTM PV adoption

- `NatLabRockies/dgen` (Python, BSD-3-Clause). Distributed generation adoption model used by ISO-NE [31].

Building simulation

- `NatLabRockies/EnergyPlus` (C++, custom (BSD-style)). Whole-building energy simulation [67].

Building-stock models

- `NatLabRockies/resstock` (Ruby, custom (BSD-style)). Residential stock model behind the EULP data set [66].
- `NatLabRockies/ComStock` (Ruby, custom (BSD-style)). Commercial stock model behind the EULP data set [66].

Agent-based grid simulation

- `gridlab-d/gridlab-d` (C++, custom (BSD-style)). GridLAB-D [68] with weather-driven thermostatic load models [69].
- `arras-energy/gridlabd` (C++, custom (BSD-style)). Arras Energy distribution of GridLAB-D.

Physics-informed models

- `maziarraissi/PINNs` (Python). Original PINN code [93].
- `lululxvi/deepxde` (Python). Maintained library for physics-informed learning.

Climate indicators

- `Ouranosinc/xclim` (Python). Climate indicators such as degree days from Ouranos (Québec), applicable to climate-conditioned weather libraries [40].

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